King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 102 Major Exam II Term 252

Original exam date: 14 April 2026. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. $\int \frac{8x}{x^3+x^2-x-1}\,dx =$

(a) $2\ln\left|\frac{x-1}{x+1}\right| - \frac{4}{x+1} + C$

(b) $\ln\left|\frac{x-1}{x+1}\right| - \frac{4}{x+1} + C$

(c) $2\ln\left|\frac{x-1}{x+1}\right| + \frac{4}{x+1} + C$

(d) $\ln\left|\frac{x-1}{x+1}\right| + \frac{4}{x+1} + C$

(e) $2\ln\left|\frac{x+1}{x-1}\right| - \frac{4}{x+1} + C$

2. If $\frac{x^2+6x+4}{x^4+8x^2+16} = \frac{Ax+B}{x^2+4} + \frac{Cx+D}{(x^2+4)^2}$, then $A+B+C+D =$

(a) 9

(b) 10

(c) 7

(d) 8

(e) 6

3. The area of the surface generated by revolving the curve $y = 2\sqrt{x}$ on the interval $[4, 9]$ about the x-axis is

(a) $\frac{4\pi}{3}\left(10\sqrt{10} + 5\sqrt{5}\right)$

(b) $\frac{4\pi}{3}\left(10\sqrt{10} - 5\sqrt{5}\right)$

(c) $\frac{2\pi}{3}\left(10\sqrt{10} - 5\sqrt{5}\right)$

(d) $\frac{8\pi}{3}\left(10\sqrt{10} - 5\sqrt{5}\right)$

(e) $\frac{8\pi}{3}\left(10\sqrt{10} + 5\sqrt{5}\right)$

4. $\int t \csc t \cot t \, dt =$

(a) $-t \csc t - 2 \ln |\csc t + \cot t| + C$

(b) $t \csc t + \ln |\csc t + \cot t| + C$

(c) $-t \csc t + \ln |\csc t + \cot t| + C$

(d) $-t \csc t - \ln |\csc t + \cot t| + C$

(e) $t \csc t - \ln |\csc t + \cot t| + C$

5. $\int \frac{d\theta}{2-3\sin\theta} =$

(a) $\frac{1}{\sqrt{5}} \ln \left| \frac{2\tan\left(\frac{\theta}{2}\right) - 3 + \sqrt{5}}{2\tan\left(\frac{\theta}{2}\right) - 3 - \sqrt{5}} \right| + C$

(b) $\ln \left| \frac{2\tan\left(\frac{\theta}{2}\right) - 3 - \sqrt{5}}{2\tan\left(\frac{\theta}{2}\right) - 3 + \sqrt{5}} \right| + C$

(c) $\ln \left| \frac{\tan\left(\frac{\theta}{2}\right) - 3 - \sqrt{5}}{\tan\left(\frac{\theta}{2}\right) - 3 + \sqrt{5}} \right| + C$

(d) $\frac{1}{\sqrt{5}} \ln \left| \frac{\tan\left(\frac{\theta}{2}\right) - 3 - \sqrt{5}}{\tan\left(\frac{\theta}{2}\right) - 3 + \sqrt{5}} \right| + C$

(e) $\frac{1}{\sqrt{5}} \ln \left| \frac{2\tan\left(\frac{\theta}{2}\right) - 3 - \sqrt{5}}{2\tan\left(\frac{\theta}{2}\right) - 3 + \sqrt{5}} \right| + C$

6. $\int \frac{d\theta}{\cos\theta - 1} =$

(a) $-\csc\theta + \cot\theta + C$

(b) $-\csc\theta - \cot\theta + C$

(c) $\csc\theta - \cot\theta + C$

(d) $\csc\theta + 2\cot\theta + C$

(e) $\csc\theta + \cot\theta + C$

7. The volume of the solid generated by revolving the region bounded by the graphs of $y = \sqrt{1-x}$, $x = 0$, $y = 0$ about the line $y = 4$ is

(a) $\frac{35\pi}{6}$

(b) $\frac{23\pi}{6}$

(c) $\frac{29\pi}{6}$

(d) $\frac{25\pi}{6}$

(e) $\frac{31\pi}{6}$

8. $\int \frac{4}{(4+x^2)^2}\,dx =$

(a) $\frac{1}{4}\left(\arctan\frac{x}{2}+\frac{2x}{x^2+4}\right)+C$

(b) $\frac{1}{2}\left(\arctan\frac{x}{2}+\frac{2x}{x^2+4}\right)+C$

(c) $\frac{1}{2}\left(\arctan\frac{x}{2}+\frac{x}{x^2+4}\right)+C$

(d) $\frac{1}{4}\left(\arctan\frac{x}{2}+\frac{x}{x^2+4}\right)+C$

(e) $\arctan\frac{x}{2}+\frac{2x}{x^2+4}+C$

9. The arc length of the graph of the function $f(x)=\ln\left(\frac{e^x+1}{e^x-1}\right)$ over the interval $[\ln 6, \ln 8]$ is

(a) $\ln\frac{17}{20}$

(b) $\ln\frac{23}{20}$

(c) $\ln\frac{29}{20}$

(d) $\ln\frac{19}{20}$

(e) $\ln\frac{27}{20}$

10. $\int \frac{x}{\sqrt{4x-x^2}}\,dx =$

(a) $2\arcsin\frac{x-2}{2}-2\sqrt{4x-x^2}+C$

(b) $\arcsin\frac{x-2}{2}+\sqrt{4x-x^2}+C$

(c) $2\arcsin\frac{x-2}{2}-\sqrt{4x-x^2}+C$

(d) $2\arcsin\frac{x-2}{2}+\sqrt{4x-x^2}+C$

(e) $\arcsin\frac{x}{2}-\sqrt{4x-x^2}+C$

11. $\int \frac{x^3 e^{x^2}}{(x^2+1)^2}\, dx =$

(a) $\dfrac{e^{x^2}}{4(x^2+1)} + C$

(b) $\dfrac{4e^{x^2}}{x^2+1} + C$

(c) $\dfrac{e^{x^2}}{x^2+1} + C$

(d) $\dfrac{e^{x^2}}{2(x^2+1)} + C$

(e) $\dfrac{2e^{x^2}}{x^2+1} + C$

12. The volume of the solid formed by revolving the region bounded by the graphs of $y = x^3$, $y = 8$, $x = 0$ about the y axis is

(a) $\dfrac{48\pi}{5}$

(b) $\dfrac{16\pi}{5}$

(c) $\dfrac{68\pi}{5}$

(d) $\dfrac{32\pi}{5}$

(e) $\dfrac{96\pi}{5}$

13. $\int \tan^5 x \sec^4 x\, dx =$

(a) $\dfrac{1}{8}\tan^8 x - \dfrac{1}{6}\tan^6 x + C$

(b) $\dfrac{1}{8}\tan^8 x + \dfrac{1}{6}\tan^6 x + C$

(c) $\dfrac{1}{7}\tan^7 x + \dfrac{1}{5}\tan^5 x + C$

(d) $\tan^8 x + \tan^6 x + C$

(e) $\dfrac{1}{7}\tan^7 x - \dfrac{1}{5}\tan^5 x + C$

14. $\int \sin^3 2\theta \sqrt{\cos 2\theta}\, d\theta =$

(a) $\dfrac{-1}{3}(\cos 2\theta)^{3/2} + \dfrac{1}{7}(\cos 2\theta)^{7/2} + C$

(b) $\dfrac{2}{3}(\cos 2\theta)^{3/2} + \dfrac{1}{7}(\cos 2\theta)^{7/2} + C$

(c) $\dfrac{-1}{3}(\cos 2\theta)^{3/2} + \dfrac{2}{7}(\cos 2\theta)^{7/2} + C$

(d) $\dfrac{1}{3}(\cos 2\theta)^{3/2} + \dfrac{1}{7}(\cos 2\theta)^{7/2} + C$

(e) $\dfrac{-1}{3}(\cos 2\theta)^{3/2} - \dfrac{1}{7}(\cos 2\theta)^{7/2} + C$

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	$\frac{8x}{(x-1)(x+1)^2} = \frac{2}{x-1} - \frac{2}{x+1} + \frac{4}{(x+1)^2}$, then integrate.
2	**(c)**	Matching coefficients gives $A = 0,\ B = 1,\ C = 6,\ D = 0$, so the sum is 7.
3	**(d)**	$S = 4\pi \int_4^9 \sqrt{x+1}\, dx = \frac{8\pi}{3}(10\sqrt{10} - 5\sqrt{5})$.
4	**(d)**	By parts, $u = t$, $v = -\csc t$: $-t\csc t + \int \csc t\, dt$.
5	**(e)**	$u = \tan\frac{\theta}{2}$: $\int \frac{du}{u^2-3u+1}$, roots $\frac{3\pm\sqrt{5}}{2}$, giving $\frac{1}{\sqrt{5}}\ln\lvert\cdot\rvert$.
6	**(e)**	Multiply by $\cos\theta + 1$: $-\csc\theta\cot\theta - \csc^2\theta$, giving $\csc\theta + \cot\theta$.
7	**(c)**	Washers: $V = \pi\int_0^1 [16 - (4 - \sqrt{1-x})^2]dx = \frac{29\pi}{6}$.
8	**(a)**	$x = 2\tan\theta$: $\frac{1}{2}\int \cos^2\theta\, d\theta = \frac{1}{4}(\theta + \sin\theta\cos\theta)$.
9	**(e)**	$\sqrt{1+f'^2} = \frac{e^x+e^{-x}}{e^x-e^{-x}}$, so $L = \ln(e^x - e^{-x})\big\rvert = \ln\frac{27}{20}$.
10	**(c)**	Complete the square, $u = x - 2$: $-\sqrt{4-u^2} + 2\arcsin\frac{u}{2}$.
11	**(d)**	$w = x^2$, then by parts: $\frac{1}{2}\int \frac{we^w}{(w+1)^2}dw = \frac{e^w}{2(w+1)}$.
12	**(e)**	Shells: $V = 2\pi\int_0^2 x(8 - x^3)\, dx = \frac{96\pi}{5}$.
13	**(b)**	$u = \tan x$: $\int (u^5 + u^7)\, du = \frac{u^6}{6} + \frac{u^8}{8}$.
14	**(a)**	$u = \cos 2\theta$: $-\frac{1}{2}\int (u^{1/2} - u^{5/2})\, du = -\frac{1}{3}u^{3/2} + \frac{1}{7}u^{7/2}$.